NICOLAS ADAMCZYK

MASTER'S STUDENT AT THE UNIVERSITÉ PARIS CITÉ
COMPUTER VISION AND AI ENGINEERING

CONTACT

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Driver's License B (own vehicle)

LINKS

GitHub - NicolasAdamczyk
LinkedIn - Nicolas Adamczyk
Website

TECHNICAL SKILLS

Vision & AI : Python (Pytorch, OpenCV)
Programming : C++, C#, Java, SQL
Digital 3D rendering : UE5, Blender, Unity
Workflow & Tools : Git, Docker, LaTeX, Excel

LANGUAGES

French (Native)
English (Advanced - Technical Reading)
Polish (Native/Bilingual)

PROFESSIONAL STRENGTHS

Scientific rigour
Technical curiosity
Team spirit & Communication

AREAS OF INTEREST

Space technologies
Motor sports (F1)
Football

ABOUT

I develop **image analysis solutions** combining **traditional processing** (OpenCV) and **deep learning** (PyTorch). I am looking for a 3-month **technical internship** (1st June - 31st August 2026) to apply my skills in **object detection** and **image analysis** (satellite, robotics or industrial). I am also proficient in **3D tools** (UE5/Blender), a major asset for synthetic vision.

PROJETS TECHNIQUES ET RÉALISATIONS

Development of a Vision System for Currency Counting and Classification
JANUARY 2026 - MARCH 2026

- Design of a **complete segmentation and classification** system **without Deep Learning** (OpenCV).
- Implementation of an **adaptive strategy** based on background (HSV/Saturation colour space) and contact management via **Hough Transform** and **Watershed**.
- **Algorithmic optimisation** reducing processing time from 60 seconds to **0.24 seconds per image**.
- Achieved **97.4% detection accuracy** (100+ images).

Development of a 3D orbital tracking system for the ISS
JULY 2025

- Development of a **3D interface** (React Three Fiber) integrating geodetic coordinate calculations.
- Processing of **TLE** (Two-Line Element) feeds via **API** for real-time orbital propagation.
- Data processing and APIs developed in **Python** (calculations), **Node.js** (back-end), **Three.js** (high-quality 3D rendering).

Development of a pharmacy management application (Team Project)
JANUARY 2025 - MAY 2025

- Design of a comprehensive **ERP system** in Java/JavaFX to manage business processes (inventory, sales, orders) via a **modular DAO architecture**.
- Implementation of a **PostgreSQL database** for managing complex relational queries and use of Git for **synchronising multi-developer integration**.
- **Recognition of the direct impact of initial design** (database schema and package structure) on performance and ease of integration of large volumes of data.

Practical application of 3D modelling
SEPTEMBRER 2022 - NOW

- Management of **photorealistic 3D scenes** in **Unreal Engine 5** and **Blender**.
- Management of **geometric complexity** (LOD, Nanite), **advanced texturing** and implementation of **dynamic lighting** (Lumen).
- Reduction of **polygon load** and **optimisation of scene rendering times**.

TRAINING & QUALIFICATIONS

Master's degree in AI Computer Vision
2025-2027
Université Paris Cité - In the first year

Junior Space Academy (Île-de-France Space Academy)
JUIN 2025
Université Paris-Saclay - Immersion in the heart of the space ecosystem
Visits to laboratories and the Paris Air Show

Bachelor's Degree in Computer Science and Applications
2022-2025
Université Paris Cité

High School Diploma
2022
Private High School L'Espérance in Aulnay-Sous-Bois
Specialising in Mathematics and Computer Science – Honours